

E-Polarized Plane-Wave Scattering by a Finite Sinusoidal PEC Grating via the Nyström Method

A. Petryshyn, S.V. Dukhopelnykov, and T.L. Zinenko
*O. Y. Usikov Institute for Radiophysics and Electronics of the
 National Academy of Sciences of Ukraine
 Kharkiv 61085, Ukraine*

Abstract—This paper presents a Nyström treatment for two-dimensional (2D) E-polarized plane-wave scattering by a finite sinusoidal grating made of a perfect electric conductor (PEC). The inverse-square-root endpoint singularity of the induced current is extracted as a factor, and the logarithmic integral equation is differentiated before the resulting Cauchy-singular equation is discretized at Gauss–Chebyshev nodes. The formulation accommodates arbitrary incidence angles. Convergence under interpolation order increment and agreement with independently implemented analytical-regularization and pulse-basis methods are demonstrated for a flat strip, whose total scattering cross section approaches the geometrical-optics limit at high frequency. Results for a symmetric five-period grating show how corrugation height and electrical size shape the absolute far-field scattering pattern. Above the first propagating-harmonic threshold, broadened finite-grating lobes appear near the reference directions of the corresponding infinite grating, while finite-length effects preserve the specular reflection lobe and end-generated background. The study therefore clarifies how infinite-grating Floquet harmonics propagation directions can be used to interpret scattering by finite gratings.

Index Terms—Electromagnetic scattering, sinusoidal open strip, integral equations, Nyström method, perfect electric conductors.

I. INTRODUCTION

Finite corrugated conductors combine periodic-surface physics with end effects that are absent from an infinite grating. Even a scalar two-dimensional model therefore has to resolve a singular edge current, interaction across several corrugations, and radiation into both halfspaces. Early numerical treatments of two-dimensional diffraction and the associated quadrature techniques are surveyed by Nazarchuk [1]. Differentiated formulations based on the method of discrete singularities (MDS) and the Nyström method for open perfect-electric-conductor (PEC) arcs, including interpolation quadratures and supplementary conditions, were developed in [2]–[5]. Related edge-conditioned and exponentially convergent Nyström schemes for open structures are reported in [6]–[8]. Accurate regularized analyses of curved open PEC reflectors provide further full-wave benchmarks [9], [10].

The scattering by PEC finite sinusoidal grating has also been studied with other methods. Kobayashi and Eizawa used a Wiener–Hopf technique analysis for the E-polarization case [11] and H-polarized analysis was reported in [12].

This work was supported by the National Research Foundation of Ukraine via project #2025.07/0094.

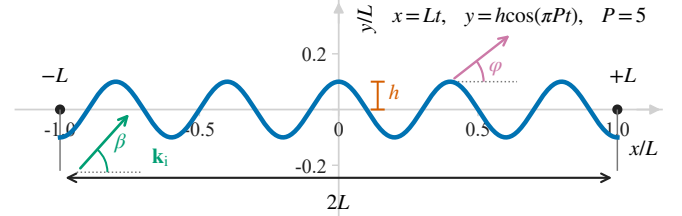


Fig. 1. Symmetric finite sinusoidal PEC grating and angle convention. The pale coordinate axes at $x = 0$ and $y = 0$ are the symmetry axis and mean line; the tick-ended orange segment denotes the corrugation height h . The incident wave vector is $\mathbf{k}_i = k\hat{\mathbf{d}}$, where $\hat{\mathbf{d}} = (\cos \beta, \sin \beta)$; observation angle φ is measured counterclockwise from $+x$.

Vinogradova, Kobayashi, and Eizawa later developed a full-wave E-polarized solution by the analytical regularization [13]. Related resonant configurations formed by two finite sinusoidal gratings were analyzed in [14]. These studies emphasize that the angles of radiation of the Floquet harmonics remain useful references, but a grating with only a few periods does not possess exact Floquet channels.

This paper presents a compact Nyström solution for the electric-field integral equation (IE) established in the scattering by a finite sinusoidal PEC grating - see Fig. 1. Its contributions are threefold. First, the grating geometry is assumed symmetric, all size parameters are referred to the half-span L , and the incidence angle is arbitrary. Second, the solution accuracy is assessed by an L^2 -norm convergence study and a triple comparison with other methods. Third, the polar scattering patterns and the total scattering cross section (TSCS) reveal the effects of corrugation height and electrical size for a five-period grating.

II. FORMULATION AND NYSTRÖM DISCRETIZATION

A. Geometry and Boundary-Value Problem

For a structure invariant along z , E polarization is represented by the scalar electric field component $U = E_z$ in the x - y plane. The scatterer is a zero-thickness PEC strip, whose cross-section Γ is symmetric in x and parameterized as

$$\Gamma(t) = (Lt, h \cos(\pi Pt)), \quad -1 \leq t \leq 1, \quad (1)$$

where $2L$ is its projected width, h/L is the relative corrugation height, and the integer P is the number of complete periods. Its projected period is $\Lambda = 2L/P$. Hence geometry is specified by

h/L and P , while frequency enters only through the electric size kL . The notation and angle convention are summarized in Fig. 1.

With the suppressed time factor $e^{-i\omega t}$, a unit-amplitude incident plane wave is

$$U^{\text{inc}}(\mathbf{r}) = \exp(ik\hat{\mathbf{d}} \cdot \mathbf{r}), \quad \hat{\mathbf{d}} = (\cos \beta, \sin \beta). \quad (2)$$

Thus constant-phase fronts travel along $\hat{\mathbf{d}}$. In computations, we use normal incidence, $\beta = \pi/2$; no such restriction is made in the equations. The scattered field U^{sc} has to satisfy the Helmholtz equation, the outgoing Sommerfeld condition, local power finiteness at both endpoints, and the Dirichlet boundary condition

$$U^{\text{inc}} + U^{\text{sc}} = 0 \quad \text{on } \Gamma. \quad (3)$$

B. Log-Singular Integral Equation

The scattered field can be presented as a convolution of the surface current density q with the Green's function (single-layer representation). As its endpoint behavior is known from the local power density finiteness condition, this density is factored as

$$q(t)|\mathbf{r}'(t)| = \frac{v(t)}{\pi\sqrt{1-t^2}}, \quad (4)$$

where v is smooth. Equation (4) extracts the singular factor; it does not remove the physical edge singularity. Define $R(t, \tau) = |\mathbf{r}(t) - \mathbf{r}(\tau)|$, $G(s) = H_0^{(1)}(ks)$, $w(t) = (1-t^2)^{-1/2}$, and $u_0(\tau) = U^{\text{inc}}(\mathbf{r}(\tau))$. Then, the boundary condition (3) yields the following logarithmically singular IE

$$\frac{1}{\pi} \int_{-1}^1 G(R(t, \tau))v(t)w(t) dt = -u_0(\tau). \quad (5)$$

Following the Nyström treatments for open PEC arcs in [2], [3], [5], the logarithmically singular IE (5) is differentiated with respect to τ , producing a Cauchy-singular IE. In particular, let

$$c_H = -\frac{2i}{\pi}, \quad (6)$$

$$K(t, \tau) = \frac{\partial}{\partial \tau} H_0^{(1)}(kR(t, \tau)) - \frac{c_H}{t - \tau}, \quad (7)$$

$$f(\tau) = -\frac{d}{d\tau} U^{\text{inc}}(\mathbf{r}(\tau)) = -ikU^{\text{inc}}(\mathbf{r}(\tau))\hat{\mathbf{d}} \cdot \mathbf{r}'(\tau). \quad (8)$$

The Cauchy-part extraction follows the normalized formulations in [2], [3]. The small-argument expansion of $H_0^{(1)}$ in [8] gives $c_H = -2i/\pi$; consequently, K is a continuous function. The differentiated equation is

$$\frac{1}{\pi} \text{p. v.} \int_{-1}^1 \left[\frac{c_H}{t - \tau} + K(t, \tau) \right] \frac{v(t)}{\sqrt{1-t^2}} dt = f(\tau). \quad (9)$$

Further, we discretize the Cauchy-singular equation (1) using the interpolation quadratures described in [2]–[4], with first-kind Gauss–Chebyshev nodes and interlaced second-kind collocation points

$$\begin{aligned} t_i &= \cos \frac{(2i-1)\pi}{2N}, & i &= 1, \dots, N, \\ \tau_j &= \cos \frac{j\pi}{N}, & j &= 1, \dots, N-1. \end{aligned} \quad (10)$$

Denoting the density samples by $v_i = v(t_i)$, we obtain the following boundary rows for $j = 1, \dots, N-1$:

$$\frac{1}{N} \sum_{i=1}^N \left[\frac{c_H}{t_i - \tau_j} + K(t_i, \tau_j) \right] v_i = f(\tau_j). \quad (11)$$

Here, the Gauss–Chebyshev weight π/N combines with the factor $1/\pi$ in (9), yielding the factor $1/N$.

As in [2], [3], a weighted supplementary condition restores the integration constant lost upon differentiation and hence equivalence to (5),

$$\frac{1}{N} \sum_{i=1}^N M(t_i)v_i = c, \quad (12)$$

where

$$M(t) = \int_{-1}^1 H_0^{(1)}(kR(t, \tau)) \frac{d\tau}{\sqrt{1-\tau^2}}, \quad (13)$$

$$c = - \int_{-1}^1 U^{\text{inc}}(\mathbf{r}(\tau)) \frac{d\tau}{\sqrt{1-\tau^2}}. \quad (14)$$

The logarithmic self term in M is subtracted analytically and its smooth remainder is evaluated by a higher-order Chebyshev rule. Equations (11) and (12) form an $N \times N$ dense system.

C. Far Field and TSCS

For $\hat{\mathbf{s}}(\varphi) = (\cos \varphi, \sin \varphi)$, we define the far-field scattering pattern Φ_{sc} by $U^{\text{sc}}(r, \varphi) \sim e^{ikr} \Phi_{\text{sc}}(\varphi)/\sqrt{kr}$. Then, the single-layer far-field asymptotics and the same quadrature in [3] give

$$\Phi_{\text{sc}}^{(N)}(\varphi) = \sqrt{\frac{2}{\pi i}} \frac{1}{N} \sum_{i=1}^N e^{-ik\hat{\mathbf{s}}(\varphi) \cdot \mathbf{r}(t_i)} v_i. \quad (15)$$

For the unit incident amplitude, the total scattering cross section is

$$\sigma = \frac{1}{k} \int_0^{2\pi} |\Phi_{\text{sc}}(\varphi)|^2 d\varphi. \quad (16)$$

It has the required length dimension. The polar plots below display the associated angular scattering pattern $10 \log_{10}[|\Phi_{\text{sc}}(\varphi)|^2/(kL)]$; they are not normalized to the maximum of each curve.

The verification, representative field pattern, height sweep, and electrical-size sweep are presented in Figs. 2–5, respectively.

For reference, it is instructive to keep in mind the plane-wave scattering by the similar infinite sinusoidal PEC grating. In this case, the scattered field is expandable in terms of the Floquet series, where the harmonics have the wavenumbers

$$\xi_m = \cos \beta + \frac{m\pi P}{kL}, \quad |\xi_m| < 1. \quad (17)$$

Their upper- and lower-halfspace angles are $\arccos \xi_m$ and $2\pi - \arccos \xi_m$, respectively. In our case of the finite grating, these angles are only the markers for the broad lobes of

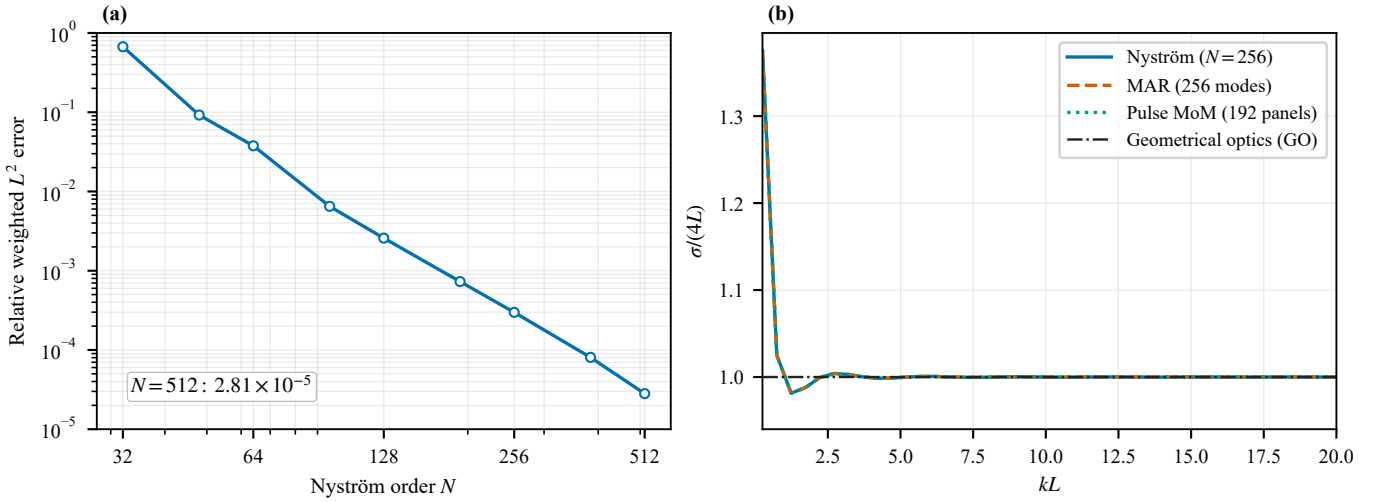


Fig. 2. Verification. (a) Relative L^2 error of the smooth density versus interpolation order for $P = 5$, $h/L = 0.10$, and $kL = 20$. (b) Flat-strip total scattering cross section (TSCS), normalized by its geometrical-optics limit $4L$, from the differentiated Nyström method ($N = 256$), the method of analytical regularization [15], [16] (256 modes, $Q = 2048$), and a pulse-basis method of moments [17], [18] (MoM; 192 elements); the dash-dotted line is the geometrical-optics (GO) limit.

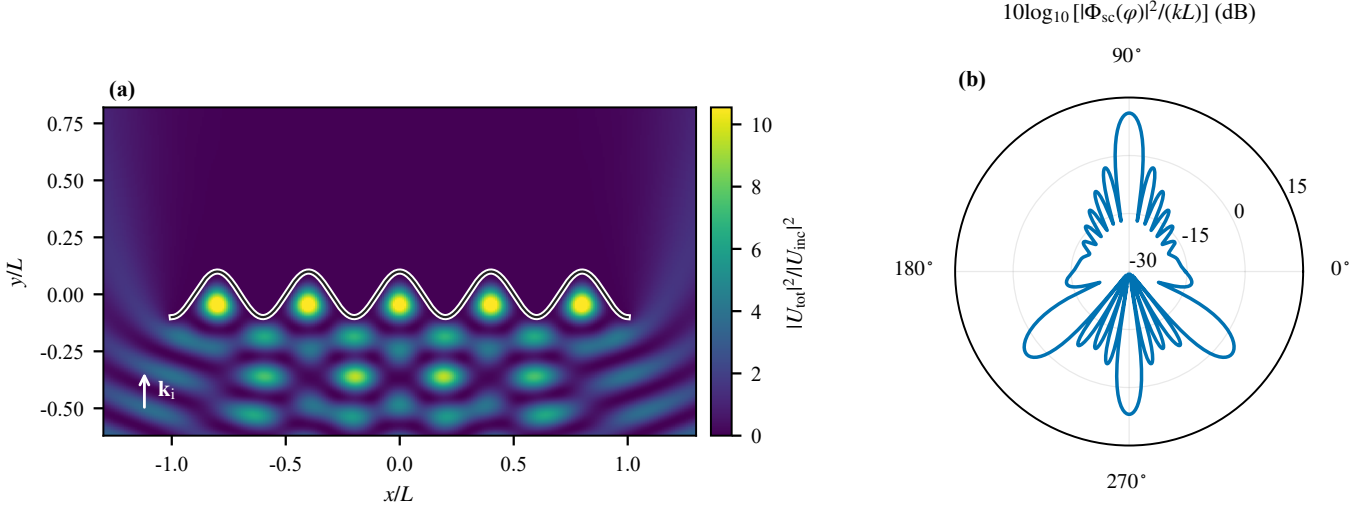


Fig. 3. Five-period grating at $P = 5$, $h/L = 0.10$, $kL = 20$, and $\beta = \pi/2$. (a) Relative total-field intensity $|U_{\text{inc}} + U^{\text{sc}}|^2/|U_{\text{inc}}|^2$. (b) Absolute polar scattering diagram based on Φ_{sc} in (15), shown as $10 \log_{10} [|\Phi_{\text{sc}}(\varphi)|^2/(kL)]$ on the common -30 to $+15$ dB scale.

the scattering pattern; below they are called infinite-grating reference directions. At the normal incidence, this reduces to

$$\cos \varphi_m = \frac{m\pi P}{kL}, \quad \left| \frac{m\pi P}{kL} \right| < 1. \quad (18)$$

For instance, if $P=5$, the first-order harmonics become propagating at

$$kL = \pi P = 15.70796 \quad (P = 5). \quad (19)$$

III. NUMERICAL RESULTS

A. Convergence and Cross-Validation

Fig. 2(a) reports a relative weighted L^2 error of the smooth part of the current density for $P = 5$, $h/L = 0.10$, $kL = 20$,

and $\beta = \pi/2$. Let $I_N v_N$ denote its Chebyshev interpolant and set $\theta_\ell = (\ell + 1/2)\pi/4096$. With the $N_{\text{ref}} = 800$ solution as the reference, the plotted error is

$$\epsilon_N = \left[\frac{\sum_{\ell=0}^{4095} |I_N v_N(\cos \theta_\ell) - I_{800} v_{800}(\cos \theta_\ell)|^2}{\sum_{\ell=0}^{4095} |I_{800} v_{800}(\cos \theta_\ell)|^2} \right]^{1/2}. \quad (20)$$

This midpoint sampling in θ approximates the $L^2_{1/\sqrt{1-t^2}}$ norm in t . For $N = 32, 48, 64, 96, 128, 192, 256, 384, 512$, the error decreases systematically and reaches 2.81142×10^{-5} at $N = 512$. The latter order is used for all corrugated-grating results below. Repeating every unique publication case at $N = 800$ changes the TSCS by at most 7.53315×10^{-6} and the full

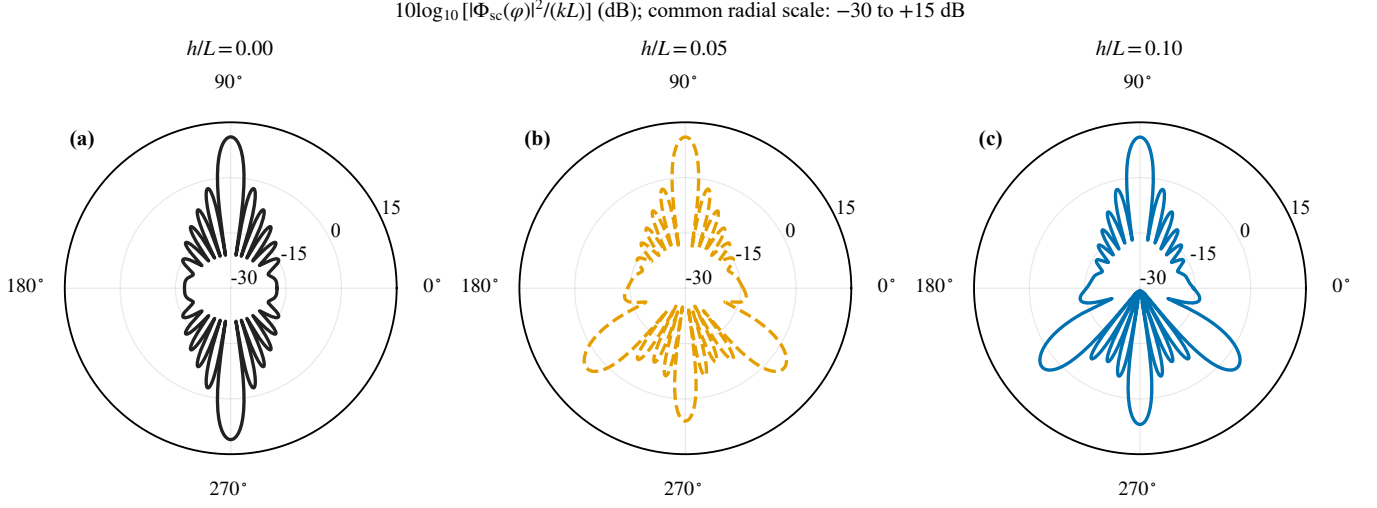


Fig. 4. Absolute polar scattering diagrams based on Φ_{sc} in (15), shown as $10\log_{10} [|\Phi_{sc}(\varphi)|^2/(kL)]$, for $P = 5$, $kL = 20$, and $\beta = \pi/2$: (a) $h/L = 0$; (b) $h/L = 0.05$; and (c) $h/L = 0.10$. Each panel contains one curve and uses the common -30 to $+15$ dB scale.

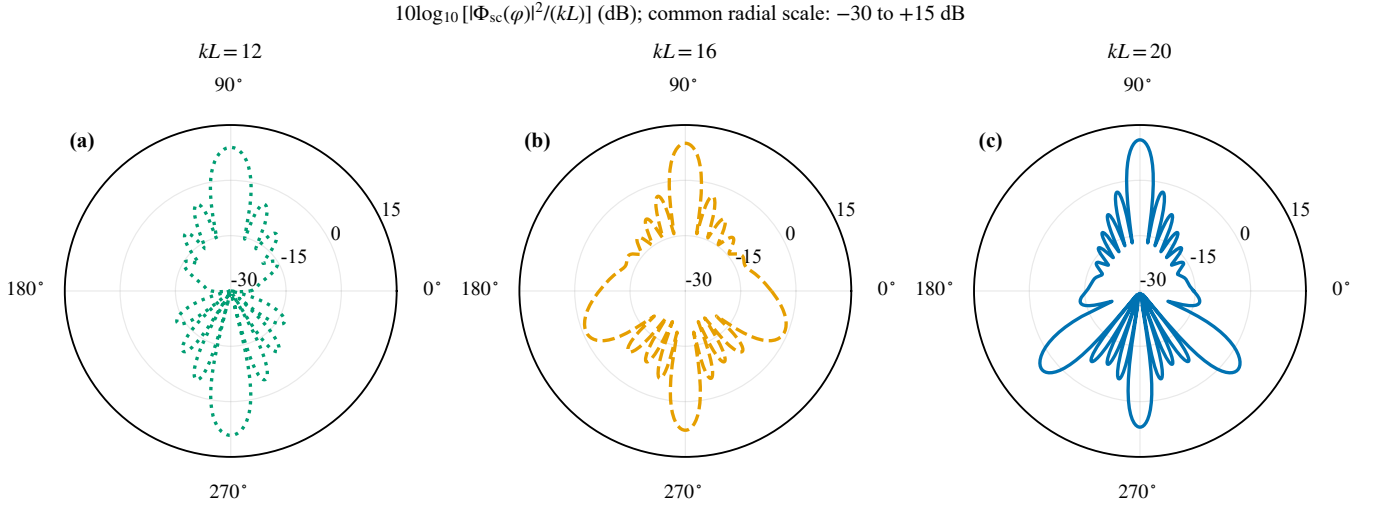


Fig. 5. Absolute polar scattering diagrams based on Φ_{sc} in (15), shown as $10\log_{10} [|\Phi_{sc}(\varphi)|^2/(kL)]$, for $P = 5$, $h/L = 0.10$, and $\beta = \pi/2$: (a) $kL = 12$; (b) $kL = 16$; and (c) $kL = 20$. Each panel contains one curve and uses the common -30 to $+15$ dB scale. The first-order periodic-reference harmonics are cut off at $kL = 5\pi$ and propagate above it.

complex far-field pattern by at most 2.30664×10^{-5} in relative L^2 norm.

The second test uses a flat strip ($h = 0$) with electrical size varying in the range $0.25 \leq kL \leq 20$. Fig. 2(b) compares the present Nyström implementation with separately obtained results of the analytical-regularization and pulse-basis method-of-moments treatments. The maximum spread in $\sigma/(4L)$ among the three results is 1.59298×10^{-5} . The first two use $N = 256$, and the pulse discretization uses 192 elements. In the most demanding scenario, at $kL = 20$, doubling those orders changes the TSCS by at most 5.03819×10^{-6} . At $kL = 20$ the Nyström value is 0.9999612. This is close to the geometrical-optics limit $\sigma/(4L) \rightarrow 1$ for a broadside-illuminated flat PEC strip of projected width $2L$ [8], but it

is correctly retained as a finite-frequency value.

B. Representative Five-Period Grating

Fig. 3 shows the results for the normal incidence for $P = 5$, $h/L = 0.10$, and $kL = 20$. As visible on panel (a), below the grating the incident and reflected fields interfere and form the standing wave pattern; above it, the pattern reveals the shadow. The polar scattering pattern on panel (b) gives Φ_{sc} in (15) over $0 \leq \varphi < 2\pi$. At this electric size, the $m = \pm 1$ periodic-reference Floquet harmonics are propagating, and the calculated finite-grating pattern contains broad off-normal lobes near the corresponding reference directions, in the reflection half-space. Despite sinusoidal corrugation, the

TABLE I

NORMALIZED TSCS FOR THE HEIGHT AND ELECTRICAL-SIZE SWEEPS IN FIGS. 4 AND 5 ($P = 5$, $\beta = \pi/2$).

$kL = 20$		$h/L = 0.10$	
h/L	$\sigma/(4L)$	kL	$\sigma/(4L)$
0	0.9999612	12	1.010974
0.05	0.9948396	16	0.9968797
0.10	0.9926829	20	0.9926829

specular reflection lobe and the edge-generated background remain because only five periods are retained.

C. Height and Electric-Size Dependence

At fixed $P = 5$ and $kL = 20$, increasing h/L changes the phase and magnitude of the induced current across each corrugation. Fig. 4 shows the resulting power redistribution among the specular and off-normal lobes. Because every panel has the same absolute radial scale, changes in level can be compared directly. The corresponding normalized TSCS values, together with those for the electrical-size sweep below, are listed in Table I.

The variation in frequency changes both the overall electric width and the set of propagating periodic-reference Floquet harmonics. The three cases for $P = 5$ and $h/L = 0.10$ in Fig. 5 are also tabulated in Table I. At $kL = 12$ the first-order periodic harmonics are evanescent. They become propagating above the threshold (19), so the first-order finite-grating lobes appear only in the $kL = 16$ pattern and scan further from the grazing directions at $kL = 20$. Their widths can be expected to get narrower if the number of periods P is taken larger.

IV. CONCLUSION

A log-singular integral equation, discretized with the Nyström technique, has been studied for the E-polarized plane-wave scattering by a finite zero-thickness sinusoidal PEC grating. We have visualized the solution convergence in the L_2 -norm and demonstrated the agreement with two other computational techniques, one of which (MAR) also has mathematically guaranteed convergence. Additionally, we have demonstrated that the frequency dependence of the broadside-illuminated flat PEC strip TSCS, computed with our code, tends to the geometrical-optics limit $4L$ and coincides with results of [8]. The operating frequency and the grating period determine when the periodic-reference Floquet harmonics become propagating, while the grating electric size (number of periods) and the corrugation height mainly influence the width and intensity of the grating lobes.

REFERENCES

- [1] Z. T. Nazarchuk, *Numerical Investigation of Wave Diffraction from Cylindrical Structures*. Kyiv, Ukraine: Naukova Dumka, 1989, in Russian.
- [2] A. A. Nosich and Yu. V. Gandel, "Method of discrete singularities in the accurate modeling of a reflector beam waveguide," in *2005 IEEE MTT-S Int. Microwave Symp. Dig.*, Long Beach, 2005, pp. 1083–1086, doi: 10.1109/MWSYM.2005.1516860.
- [3] A. A. Nosich and Yu. V. Gandel, "Numerical analysis of quasioptical multireflector antennas in 2-D with the method of discrete singularities: E-wave case," *IEEE Trans. Antennas Propag.*, vol. 55, no. 2, pp. 399–406, Feb. 2007, doi: 10.1109/TAP.2006.889811.
- [4] A. A. Nosich, Yu. V. Gandel, T. Magath, and A. Altintas, "Numerical analysis and synthesis of 2D quasi-optical reflectors and beam waveguides based on an integral-equation approach with Nyström's discretization," *J. Opt. Soc. Amer. A*, vol. 24, no. 9, pp. 2831–2836, Sep. 2007, doi: 10.1364/JOSAA.24.002831.
- [5] Yu. V. Gandel, "Boundary-value problems for the Helmholtz equation and their discrete mathematical models," *J. Math. Sci.*, vol. 171, no. 1, pp. 74–88, 2010, doi: 10.1007/s10958-010-0127-3.
- [6] M. S. Tong and W. C. Chew, "Nyström method with edge condition for electromagnetic scattering by 2D open structures," *Prog. Electromagn. Res.*, vol. 62, pp. 49–68, 2006, doi: 10.2528/PIER06021901.
- [7] J. L. Tsalamengas, "Exponentially converging Nyström's methods for systems of singular integral equations with applications to open/closed strip- or slot-loaded 2-D structures," *IEEE Trans. Antennas Propag.*, vol. 54, no. 5, pp. 1549–1558, May 2006, doi: 10.1109/TAP.2006.874348.
- [8] O. V. Shapoval, R. Sauleau, and A. I. Nosich, "Scattering and absorption of waves by flat material strips analyzed using generalized boundary conditions and Nyström-type algorithm," *IEEE Trans. Antennas Propag.*, vol. 59, no. 9, pp. 3339–3346, Sep. 2011, doi: 10.1109/TAP.2011.2161547.
- [9] T. Oğuzer, A. I. Nosich, and A. Altıntaş, "E-polarized beam scattering by an open cylindrical PEC strip having an arbitrary "conical-section" profile," *Microw. Opt. Technol. Lett.*, vol. 31, no. 6, pp. 480–484, Dec. 2001, doi: 10.1002/mop.10067.
- [10] T. Oğuzer, A. Altıntaş, and A. I. Nosich, "Integral equation analysis of an arbitrary-profile and varying-resistivity cylindrical reflector illuminated by an E-polarized complex-source-point beam," *J. Opt. Soc. America A*, vol. 26, no. 7, pp. 1525–1532, 2009, doi: 10.1364/JOSAA.26.001525.
- [11] K. Kobayashi and T. Eizawa, "Plane wave diffraction by a finite sinusoidal grating," *IEICE Trans. Electron.*, vol. E74-C, no. 9, pp. 2815–2826, Sep. 1991, doi: 10.1587/e74-c_9_2815.
- [12] T. Eizawa and K. Kobayashi, "Wiener-Hopf analysis of the H-polarized plane wave diffraction by a finite sinusoidal grating," *Prog. Electromagn. Res.*, vol. 149, pp. 1–13, 2014, doi: 10.2528/PIER14063007.
- [13] E. D. Vinogradova, K. Kobayashi, and T. Eizawa, "Full wave analysis of plane wave diffraction by a finite sinusoidal grating: E-polarization case," *Wave Motion*, vol. 86, pp. 44–62, Mar. 2019, doi: 10.1016/j.wavemoti.2018.12.006.
- [14] E. D. Vinogradova and K. Kobayashi, "Complex eigenvalues of natural TM-oscillations in an open resonator formed by two sinusoidally corrugated metallic strips," *IET Microw., Antennas Propag.*, vol. 15, no. 10, pp. 1283–1298, 2021, doi: 10.1049/mia.2.12166.
- [15] A. I. Nosich, "The method of analytical regularization in wave-scattering and eigenvalue problems: Foundations and review of solutions," *IEEE Antennas Propag. Mag.*, vol. 41, no. 3, pp. 34–49, Jun. 1999, doi: 10.1109/74.775246.
- [16] A. I. Nosich, "Method of analytical regularization in computational photonics," *Radio Sci.*, vol. 51, no. 8, pp. 1421–1430, Aug. 2016, doi: 10.1002/2016RS006044.
- [17] R. F. Harrington, "Matrix methods for field problems," *Proc. IEEE*, vol. 55, no. 2, pp. 136–149, Feb. 1967, doi: 10.1109/PROC.1967.5433.
- [18] S. Hatamzadeh-Varmazyar, M. Naser-Moghadasi, and Z. Masouri, "A moment method simulation of electromagnetic scattering from conducting bodies," *Prog. Electromagn. Res.*, vol. 81, pp. 99–119, 2008, doi: 10.2528/PIER07122502.